

DBMS Unit 5

DBMS UNIT–V Question Bank Categorized by Topics

1. Transaction Concept

1.1 Transaction Concept / Properties / States

Q1 — C05 (07)

What is meant by State Transition Diagram? When is a transaction said to be failed? Explain with a diagram.

Q2 — C05 (07)

With a neat diagram, write and explain the state transition diagram illustrating the states for transaction execution.

Q3 — C05 (07)

With a neat diagram, write and explain the state transition diagram illustrating the states for transaction execution.

Q4 — C05 (07)

Explain properties of transaction with state transaction diagram.

Q5 — C05 (08)

Illustrate ACID Properties in Transaction. Discuss the isolation levels of transaction.

Q6 — C05 (07)

Explain ACID properties in Transaction.

Q7 — C05 (08)

What do you understand by a transaction? In what situation a transaction is said to be committed or aborted?

1.2 Simple Transaction Model / Read-Write Operations

Q8 — C05 (06)

Discuss the actions taken by the read_item and write_item operations on a database.

Q9 — C05 (06)

Discuss the actions taken by the read item and write item operations on a database.

Q10 — C05 (06)

Consider a database with objects X and Y and assume that there are two transactions T1 and T2.

Transaction T1 reads objects X and Y and then writes object X.

Transaction T2 reads objects X and Y and then writes objects X and Y.

- i. Give an example schedule with actions of transactions T1 and T2 on objects X and Y that results in a write-read conflict.
 - ii. Give an example schedule with actions of transactions T1 and T2 on objects X and Y that results in a read-write conflict.
 - iii. Give an example schedule with actions of transactions T1 and T2 on objects X and Y that results in a write-write conflict.
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2. Serializability

2.1 Conflict Serializability / Precedence Graphs

Q11 — C05 (06)

For each of the following schedules, state whether they are serializable and whether they are conflict-serializable. For each schedule, draw the corresponding

precedence graph. If the schedule is conflict-serializable, show all the conflict-equivalent serial schedules.

- i. $S_1 = r_1(A) r_2(A) w_1(A) w_2(A)$
- ii. $S_2 = r_1(A) r_2(B) w_3(A) r_2(A) r_1(B)$
- iii. $S_3 = r_1(A) w_2(A) w_1(A) r_3(A)$

Q12 — CO5 (08)

Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules.

- i. $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$
- ii. $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$
- iii. $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$
- iv. $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$

Q13 — CO5 (06)

Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. Draw the precedence graph for each.

- i. $r_1(x); r_2(z); r_1(x); r_3(x); r_3(y); w_1(x); w_3(y); r_2(y); w_2(z); w_2(y)$
- ii. $r_1(x); r_2(z); r_3(x); r_1(z); r_2(y); r_3(y); w_1(x); w_2(z); w_3(y); w_2(y)$

Q14 — CO5 (06)

Consider the three transactions T_1 , T_2 , and T_3 , and the schedules S_1 and S_2 given below. Draw the serializability (precedence) graphs for S_1 and S_2 , and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s).

$T_1: r_1(X); r_1(Z); w_1(X);$

$T_2: r_2(Z); r_2(Y); w_2(Z); w_2(Y);$

$T_3: r_3(X); r_3(Y); w_3(Y);$

$S_1: r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); w_3(Y); r_2(Y); w_2(Z); w_2(Y);$

S2: r1(X); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); w1(X); w2(Z); w3(Y); w2(Y);

Q15 — C05 (10)

Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules.

- i. r1(X); r3(X); w1(X); r2(X); w3(X);
- ii. r1(X); r3(X); w3(X); w1(X); r2(X);
- iii. r3(X); r2(X); w3(X); r1(X); w1(X);
- iv. r3(X); r2(X); r1(X); w3(X); w1(X);

Q16 — C05 (10)

Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules.

- a. r1(X); r3(X); w1(X); r2(X); w3(X);
- b. r1(X); r3(X); w3(X); w1(X); r2(X);
- c. r3(X); r2(X); w3(X); r1(X); w1(X);
- d. r3(X); r2(X); r1(X); w3(X); w1(X)

Q17 — C05 (10)

Write an Algorithm for testing conflict serializability of a schedule S with an example.

Q18 — C05 (06)

Explain how serializability is used to enforce concurrency control in database System.

3. Transaction Isolation and Atomicity

3.1 Isolation Levels / Concurrency Problems

Q19 — C05 (08)

Illustrate ACID Properties in Transaction. Discuss the isolation levels of transaction.

Q20 — C05 (09)

Why concurrency control is needed? Discuss the types of problems that might be encountered if transactions are allowed to run concurrently.

Q21 — C05 (08)

Why is concurrency control needed? Discuss the types of problems that might be encountered if transactions are allowed to run concurrently.

Q22 — C05 (07)

What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples.

Q23 — C05 (10)

What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples.

Q24 — C05 (06)

What is the Need of concurrency control and recovery?

Q25 — C05 (05)

Discuss on The Incorrect Summary Problem occurs when the transaction is executed concurrently.

4. Concurrency Control

4.1 Lock-Based Protocols / Two Phase Locking

Q26 — C05 (07)

Describe the procedure of two phase locking? Will two phase locking result in serializable schedule? Will two phase locking result in deadlock? Demonstrate

with an example.

Q27 — C05 (10)

What is two phase locking? Will two phase locking result in Serializable schedule? Will two phase locking result in deadlock? Demonstrate with an example.

Q28 — C05 (06)

What is two phase protocol? How does it guarantee serializability?

Q29 — C05 (06)

What is the two-phase locking protocol? How does it guarantee serializability?

Q30 — C05 (06)

Write a note on binary locks and explain with an example.

Q31 — C05 (04)

What is a timestamp? How does the system generate timestamps?

4.2 Deadlock Handling

Q32 — C05 (07)

Discuss deadlock prevention protocols.

Q33 — C05 (07)

Explain the process of starvation and detection techniques.

5. Recovery System

5.1 Failure Classification / Recovery and Atomicity

Q34 — C05 (07)

What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important?

Q35 — CO5 (08)

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5.2 Recovery Algorithms / Deferred & Immediate Update / ARIES

Q36 — CO5 (06)

Explain ARIES recovery algorithm with an example.

Q37 — CO5 (08)

What is ARIES? Discuss its three phases. On what principles is it based on?

Q38 — CO5 (05)

Write the procedure for recovery based on immediate update in multiple user environment.

Q39 — CO5 (06)

Write the procedure for recovery based on immediate update in multiple user environment.

Q40 — CO5 (06)

Explain the recovery techniques based on deferred update.

Q41 — CO5 (08)

Discuss Recovery techniques based on Deferred Update.

Q42 — CO5 (07)

Discuss shadow paging. How it is different from log based recovery techniques? Mention some of its disadvantages.